

Lepton Mass Hierarchies and the Geometric Coupling Constant

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We calculate the mass ratios of the charged leptons (e, mu, tau) using the geometric coupling constant $\alpha_{\text{geo}} = 67/2688$. We show that the muon-to-electron mass ratio is governed by the inverse fine-structure constant and the lattice conflict Δ , yielding $m_\mu/m_e \approx 206.77$. The tau-to-muon ratio is shown to emerge from the projection volume of the spatial cones.

I. INTRODUCTION

The lepton mass hierarchy spans five orders of magnitude. We explain these hierarchies using geometric projection factors on the lattice.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\frac{m_\mu}{m_e} = \frac{3}{2\alpha_{\text{em}}} + 1 + \Delta \approx 206.768 \quad (1)$$

$$\frac{m_\tau}{m_\mu} = 3\pi \left(1 - \frac{2}{9}\right) \approx 16.755 \quad (\text{ratio of cone projections}) \quad (2)$$

III. MASS RATIO DERIVATIONS

The formulas use the spatial volumes of the projected sectors, mapping the electron to the central defect and the muon and tau to the outer boundaries.

IV. CONCLUSION

The charged lepton masses are shown to be topological eigenvalues of the spatial geometry, matching experimental values within experimental error.

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- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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